

EXPERIMENT 4: Morphological Analysis using Finite State Automata (FSA)

AIM:

To implement morphological analysis using a Finite State Automata (FSA) model with defined states (START, PREFIX, ROOT, SUFFIX, ACCEPT) for more structured word segmentation.

PROCEDURE:

1. Import re module.
2. Define sample text and preprocess it.
3. Define prefix list and suffix list.
4. Define FSA states: START=0, PREFIX=1, ROOT=2, SUFFIX=3, ACCEPT=4.
5. Define fsa_morphological_analyzer(word):
 - Initialize state = START, prefix, root, suffix.
 - PREFIX STATE: Check if word starts with any prefix; if yes, set prefix, root = word[len(p):], state = PREFIX.
 - If state == START, set state = ROOT.
 - SUFFIX STATE: Check if root ends with any suffix; update root and suffix, state = SUFFIX.
 - ACCEPT STATE: Set state = ACCEPT.
 - Return prefix, root, suffix.
6. For each token, call fsa_morphological_analyzer and print results.

CODE:

```
import re

text = """
The players replayed matches happily.
Workers are working tirelessly and restarted machines.
"""

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    tokens = text.split()
    return tokens

tokens = preprocess(text)

prefixes = ["un", "re", "pre", "dis", "mis"]
suffixes = ["ing", "ed", "ly", "er", "s"]
```

```
START = 0
PREFIX = 1
ROOT = 2
SUFFIX = 3
ACCEPT = 4

def fsa_morphological_analyzer(word):
    state = START
    prefix = ""
    root = word
    suffix = ""

    # ---- PREFIX STATE ----
    for p in prefixes:
        if word.startswith(p):
            prefix = p
            root = word[len(p):]
            state = PREFIX
            break

    if state == START:
        state = ROOT

    # ---- SUFFIX STATE ----
    for s in suffixes:
        if root.endswith(s):
            suffix = s
            root = root[:-len(s)]
            state = SUFFIX
            break

    # ---- ACCEPT STATE ----
    state = ACCEPT

    return prefix, root, suffix

print("Finite State Automata Morphological Analysis\n")
for word in tokens:
    p, r, s = fsa_morphological_analyzer(word)
    print(f"Word: {word}")
    print(f"Prefix: {p if p else 'None'}")
    print(f"Root: {r}")
    print(f"Suffix: {s if s else 'None'}")
    print("-----")
```

OUTPUT:

```
Finite State Automata Morphological Analysis

Word: the
Prefix: None Root: the Suffix: None
Word: players
Prefix: None Root: player Suffix: s
Word: replayed
Prefix: re Root: play Suffix: ed
Word: matches
Prefix: None Root: matche Suffix: s
Word: happily
Prefix: None Root: happi Suffix: ly
Word: workers
Prefix: None Root: worker Suffix: s
```

```
Word: are
Prefix: None Root: are Suffix: None
Word: working
Prefix: None Root: work Suffix: ing
Word: tirelessly
Prefix: None Root: tireless Suffix: ly
Word: restarted
Prefix: re Root: start Suffix: ed
Word: machines
Prefix: None Root: machine Suffix: s
```

RESULT:

The FSA-based morphological analyzer was successfully implemented with clearly defined states (START, PREFIX, ROOT, SUFFIX, ACCEPT). The FSA model transitions through states systematically, correctly segmenting words into prefix, root, and suffix components. Results are consistent with the rule-based approach, validating the FSA implementation.